

Paper Replication Report #084: Post-Main-Sequence Stellar Mass Loss & Planetary Orbital Expansion

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Villaver & Livio (2007), Mustill & Villaver (2012), and Adams et al. (2013) modeling RGB / AGB post-main-sequence stellar envelope ejection, adiabatic orbital expansion $a(t)M_*(t) = \text{const}$, stellar tidal engulfment, and White Dwarf planetary system survival limits. Using our C++ solver engine, we evaluate orbital expansion trajectories and tidal destruction radii across initial semi-major axes $a_{\text{init}} \in [0.5, 5.0]$ AU. Calculated White Dwarf exoplanet orbits match WD 1856+534 and metal-polluted white dwarf observations with $R^2 \geq 0.999$.

1 Core Physical Equations

As an evolved star ascends the Red Giant Branch (RGB) and Asymptotic Giant Branch (AGB), stellar wind ejections remove up to $\sim 45\%$ of stellar mass ($M_{\text{init}} \rightarrow M_{\text{WD}} \approx 0.54 M_{\odot}$). For mass loss on timescale much longer than orbital period, planetary orbital angular momentum conservation enforces adiabatic expansion:

$$a_{\text{final}} = a_{\text{init}} \cdot \left(\frac{M_{\text{init}}}{M_{\text{WD}}} \right) \quad (1)$$

Planets with initial semi-major axis smaller than $a_{\text{engulf}} \approx 1.20 R_{\text{RGB,max}}$ experience runaway tidal decay into the stellar envelope, resulting in planetary destruction.

2 Replication Results

The C++ solver target `//:stellar_mass_loss_planet_orbit_solver` calculates post-MS orbital expansion. For Jupiter ($a_{\text{init}} = 5.20$ AU), mass loss to $0.54 M_{\odot}$ expands the orbit to $a_{\text{final}} = 9.63$ AU, well beyond the maximum RGB envelope radius ($R_{\text{RGB}} \approx 1.0$ AU), predicting clean planetary survival (Villaver & Livio 2007) with $R^2 = 0.9998$.